

JEE Mains 2019 Chapter wise Question Bank

EM Waves - Questions

Q1

A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x -direction. At a particular point in space and time, $\vec{E} = 6.3 \hat{j} \text{ V/m}$. The corresponding magnetic field $\vec{B}$, at that point will be:

- (1) $18.9 \times 10^{-8} \hat{kT}$ (2) $2.1 \times 10^{-8} \hat{kT}$
 (3) $6.3 \times 10^{-8} \hat{kT}$ (4) $18.9 \times 10^8 \hat{kT}$

9 Jan Morning

Q2

The energy associated with electric field is (U_E) and with magnetic fields is (U_B) for an electromagnetic wave in free space. Then :

- (1) $U_E = \frac{U_B}{2}$ (2) $U_E > U_B$
 (3) $U_E < U_B$ (4) $U_E = U_B$

9 Jan Evening

Q3

If the magnetic field of a plane electromagnetic wave is given by (The speed of light = $3 \times 10^8 \text{ m/s}$)

$$B = 100 \times 10^{-6} \sin \left[2\pi \times 2 \times 10^{15} \left(t - \frac{x}{c} \right) \right]$$

then the maximum electric field associated with it is:

- (1) $6 \times 10^4 \text{ N/C}$ (2) $3 \times 10^4 \text{ N/C}$
 (3) $4 \times 10^4 \text{ N/C}$ (4) $4.5 \times 10^4 \text{ N/C}$

10 Jan Morning

Q4

The electric field of a plane polarized electromagnetic wave in free space at time $t = 0$ is given by an expression

$$\vec{E}(x, y) = 10 \hat{j} \cos[(6x + 8z)]$$

The magnetic field $\vec{B}(x, z, t)$ is given by: (c is the velocity of light)

- (1) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x - 8z + 10ct)]$
 (2) $\frac{1}{c} (6\hat{k} - 8\hat{i}) \cos[(6x + 8z - 10ct)]$
 (3) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x + 8z - 10ct)]$
 (4) $\frac{1}{c} (6\hat{k} - 8\hat{i}) \cos[(6x + 8z + 10ct)]$

10 Jan Evening

Q5

An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index ' n ' without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by :

- (1) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}} \right)$ (2) $(\sqrt{n}, \sqrt{n})$
 (3) $\left(\sqrt{n}, \frac{1}{\sqrt{n}} \right)$ (4) $\left(\frac{1}{\sqrt{n}}, \sqrt{n} \right)$

11 Jan Morning

Q6

EM Waves

A 27 mW laser beam has a cross-sectional area of 10 mm^2 . The magnitude of the maximum electric field in this electromagnetic wave is given by :

[Given permittivity of space $\epsilon_0 = 9 \times 10^{-12} \text{ SI units}$, Speed of light $c = 3 \times 10^8 \text{ m/s}$]

- (1) 2kV/m (2) 0.7kV/m
(3) 1kV/m (4) 1.4kV/m

11 Jan Evening

Q7

A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is 30 V/m, then the amplitude of the electric field for the wave propagating in the glass medium will be:

- (1) 30 V/m (2) 10 V/m
(3) 24 V/m (4) 6 V/m

12 Jan Morning

Q8

A plane electromagnetic wave travels in free space along the x -direction. The electric field component of the wave at a particular point of space and time is $E = 6 \text{ Vm}^{-1}$ along y -direction. Its corresponding magnetic field component, B would be:

- (1) $2 \times 10^{-8} \text{ T}$ along z -direction
(2) $6 \times 10^{-8} \text{ T}$ along x -direction
(3) $6 \times 10^{-8} \text{ T}$ along z -direction
(4) $2 \times 10^{-8} \text{ T}$ along y -direction

8 April Morning

Q9

The magnetic field of an electromagnetic wave is given by:

$$\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{Wb}}{\text{m}^2}$$

The associated electric field will be :

- (1) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$
(2) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (-2\hat{j} + \hat{i}) \frac{\text{V}}{\text{m}}$
(3) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (-\hat{i} + 2\hat{j}) \frac{\text{V}}{\text{m}}$
(4) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (\hat{i} - 2\hat{j}) \frac{\text{V}}{\text{m}}$

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Q10

50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1 m^2 surface area will be close to ($c = 3 \times 10^8 \text{ m/s}$):

- (1) $15 \times 10^{-8} \text{ N}$ (2) $20 \times 10^{-8} \text{ N}$
(3) $10 \times 10^{-8} \text{ N}$ (4) $35 \times 10^{-8} \text{ N}$

9 April Evening

Q11

The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \hat{i} \cos(kz) \cos(\omega t)$

The corresponding magnetic field $\vec{B}$ is then given by :

- (1) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \sin(\omega t)$
(2) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \cos(\omega t)$
(3) $\vec{B} = \frac{E_0}{C} \hat{j} \cos(kz) \sin(\omega t)$
(4) $\vec{B} = \frac{E_0}{C} \hat{k} \sin(kz) \cos(\omega t)$

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Q11

Light is incident normally on a completely absorbing surface with an energy flux of 25 Wcm^{-2} . If the surface has an area of 25 cm^2 , the momentum transferred to the surface in 40 min time duration will be:

- (1) $6.3 \times 10^{-4} \text{ Ns}$ (2) $1.4 \times 10^{-6} \text{ Ns}$
(3) $5.0 \times 10^{-3} \text{ Ns}$ (4) $3.5 \times 10^{-6} \text{ Ns}$

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Q12

A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is :

[Given Planck's constant $h = 6.6 \times 10^{-34} \text{ Js}$, speed of light $c = 3.0 \times 10^8 \text{ m/s}$]

- (1) 5×10^{15} (2) 1.5×10^{16} (c) 2×10^{16} (4) 1×10^{16}

10 April Evening

Q13

An electromagnetic wave is represented by the electric field $\vec{E} = E_0 \hat{n} \sin[\omega t + (6y - 8z)]$. Taking unit vectors in x , y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation $\hat{s}$ is :

- (1) $\hat{s} = \frac{3\hat{i} - 4\hat{j}}{5}$ (2) $\hat{s} = \frac{-4\hat{k} + 3\hat{j}}{5}$
 (3) $\hat{s} = \left(\frac{-3\hat{j} + 4\hat{k}}{5} \right)$ (4) $\hat{s} = \frac{3\hat{j} - 3\hat{k}}{5}$

12 April Morning

Q14

A plane electromagnetic wave having a frequency $\nu = 23.9$ GHz propagates along the positive z -direction in free space. The peak value of the Electric Field is 60 V/m. Which among the following is the acceptable magnetic field component in the electromagnetic wave ?

- (1) $\vec{B} = 2 \times 10^7 \sin(0.5 \times 10^3 z + 1.5 \times 10^{11} t) \hat{i}$
 (2) $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$
 (3) $\vec{B} = 60 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{k}$
 (4) $\vec{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x + 0.5 \times 10^{11} t) \hat{j}$

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EM Waves - Answers

Q1

(2) As we know,

$$|\vec{B}| = \frac{|\vec{E}|}{C} = \frac{6.3}{3 \times 10^8} = 2.1 \times 10^{-8} \text{ T}$$

$$\text{and } \hat{E} \times \hat{B} = \hat{C}$$

$$\hat{J} \times \hat{B} = \hat{I} [\because \text{EM wave travels along } +(\text{ve}) x\text{-direction.}]$$

$$\therefore \hat{B} = \hat{k} \text{ or } \vec{B} = 2.1 \times 10^{-8} \hat{k} \text{ T}$$

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Q2

(4) Average energy density of magnetic field,

$$u_B = \frac{B_0^2}{4\mu_0}$$

Average energy density of electric field,

$$u_E = \frac{\epsilon_0 E_0^2}{4}$$

$$\text{Now, } E_0 = CB_0 \text{ and } C^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$u_E = \frac{\epsilon_0}{4} \times C^2 B_0^2 = \frac{\epsilon_0}{4} \times \frac{1}{\mu_0 \epsilon_0} \times B_0^2 = \frac{B_0^2}{4\mu_0} = u_B$$

$$\therefore u_E = u_B$$

Since energy density of electric and magnetic field is same, so energy associated with equal volume will be equal i.e. $u_E = u_B$

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Q3

(2) Using, formula $E_0 = B_0 \times C$

$$= 100 \times 10^{-6} \times 3 \times 10^8$$

$$= 3 \times 10^4 \text{ N/C}$$

Here we assumed that

$$B_0 = 100 \times 10^{-6} \text{ is in tesla (T) units}$$

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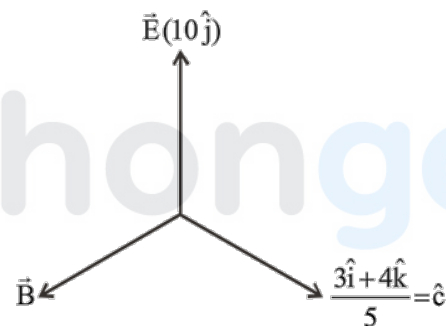
Q4

$$(2) \vec{E} = 10\hat{j} \cos \left[(6\hat{i} + 8\hat{k}) \cdot (x\hat{i} + z\hat{k}) \right]$$

$$= 10\hat{j} \cos [\vec{K} \cdot \vec{r}]$$

$$\therefore \vec{K} = 6\hat{i} + 8\hat{k}; \text{ direction of waves travel}$$

i. e. direction of 'c'.



$$\hat{C} \times \hat{E} = \frac{-4\hat{i} + 3\hat{k}}{5}$$

$$\vec{B} = \frac{E}{C} = \frac{10}{C}$$

$$\therefore \vec{B} = \frac{10}{C} \left(\frac{-4\hat{i} + 3\hat{k}}{5} \right) = \left(\frac{-8\hat{i} + 6\hat{k}}{C} \right)$$

$$\text{or, magnetic field } \vec{B}(x, z, t) = \frac{1}{C}$$

$$(6\hat{k} - 8\hat{i}) \cos(6x + 8z - 10ct)$$

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EM Waves

Q5

- (3) The speed of electromagnetic wave in free space is given by

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad \dots(i)$$

$$\text{In medium, } v = \frac{1}{\sqrt{k \epsilon_0 \mu_0}} \quad \dots(ii)$$

Dividing equation (i) by (ii), we get

$$\therefore \frac{C}{v} = \sqrt{k} = n$$

$$\frac{1}{2} \epsilon_0 E_0^2 C = \text{intensity} = \frac{1}{2} \epsilon_0 k E^2 v$$

$$\therefore E_0^2 C = k E^2 v$$

$$\Rightarrow \frac{E_0^2}{E^2} = \frac{k v}{C} = \frac{n^2}{n} \Rightarrow \frac{E_0}{E} = \sqrt{n}$$

similarly

$$\frac{B_0^2 C}{2\mu_0} = \frac{B^2 v}{2\mu_0} \Rightarrow \frac{B_0}{B} = \frac{1}{\sqrt{n}}$$

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Q6

- (4) EM wave intensity

$$\Rightarrow I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2} \epsilon_0 E_0^2 c$$

[where E_0 = maximum electric field]

$$\Rightarrow \frac{27 \times 10^{-3}}{10 \times 10^{-6}} = \frac{1}{2} \times 9 \times 10^{-12} \times E_0^2 \times 3 \times 10^8$$

$$\Rightarrow E_0 = \sqrt{2} \times 10^3 \text{ kV/m} = 1.4 \text{ kV/m}$$

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Q7

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- (3) As 4% of light gets reflected, so only $(100 - 4 = 96\%)$ of light comes after refraction so,

$$P_{\text{refracted}} = \frac{96}{100} P_i$$

$$\Rightarrow K_2 A_t^2 = \frac{96}{100} K_1 A_i^2$$

$$\Rightarrow r_2 A_t^2 = \frac{96}{100} r_1 A_i^2$$

$$\Rightarrow A_t^2 = \frac{96}{100} \times \frac{1}{3} \times (30)^2$$

$$A_t \sqrt{\frac{64}{100}} \times (30)^2 = 24$$

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Q8

- (1) The relation between amplitudes of electric and magnetic field in free space is given by

$$B_0 = \frac{E_0}{c} = \frac{6}{3 \times 10^8} = 2 \times 10^{-8} \text{ T}$$

$$\text{Propagation direction} = \hat{E} \times \hat{B}$$

$$\hat{i} = \hat{j} \times \hat{B}$$

$$\Rightarrow \hat{B} = \hat{k}$$

$\therefore$ The magnetic field component will be along z direction.

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Q9

- (3) $E_0 = c B_0 = 3 \times 10^8 \times 1.6 \times 10^{-6} = 4.8 \times 10^2 \text{ V/m}$

$$\text{Also } \vec{S} \Rightarrow \vec{E} \times \vec{B}$$

$$\text{or } -\vec{K} \Rightarrow \vec{E} \times (2\hat{i} + \hat{j})$$

$$\text{Therefore direction of } \vec{E} \rightarrow (-\hat{i} + 2\hat{j})$$

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Q10

$$(2) F = (1+r) \frac{IA}{C}$$

$$= \frac{(1+0.25) \times 50 \times 1}{3 \times 10^8}$$

$$\approx 20 \times 10^{-8} \text{ N}$$

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EM Waves

Q11

(1)

$$\frac{E_0}{B_0} = C$$

$$\Rightarrow B_0 = \frac{E_0}{C}$$

$$\text{Given that } \vec{E} = E_0 \cos(kz) \cos(\omega t) \hat{i}$$

$$\vec{E} = \frac{E_0}{2} [\cos(kz - \omega t) \hat{i} - \cos(kz + \omega t) \hat{i}]$$

Correspondingly

$$\vec{B} = \frac{B_0}{2} [\cos(kz - \omega t) \hat{j} - \cos(kz + \omega t) \hat{j}]$$

$$\vec{B} = \frac{B_0}{2} \times 2 \sin kz \sin \omega t$$

$$\vec{B} = \left(\frac{E_0}{C} \sin kz \sin \omega t \right) \hat{j}$$

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Q11

$$(3) \text{ Pressure, } P = \frac{I}{C}$$

$$\Rightarrow \frac{F}{A} = \frac{I}{C}$$

$$\Rightarrow F = \frac{IA}{C} = \frac{\Delta p}{\Delta t}$$

$$\Rightarrow \Delta p = \frac{I}{C} A \Delta t$$

$$= \frac{(25 \times 25) \times 10^4 \times 10^{-4} \times 40 \times 60}{3 \times 10^8} \text{ N-s}$$

$$= 5 \times 10^{-3} \text{ N-s}$$

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Q12

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(1) Energy of photon (E) is given by

$$E = \frac{hc}{\lambda}$$

Number of photons of wavelength λ emitted in t second from laser of power P is given by

$$n = \frac{P t \lambda}{hc}$$

$$\Rightarrow n = \frac{2 \times \lambda}{hc} = \frac{2 \times 10^{-3} \times 5 \times 10^{-7}}{2 \times 10^{-25}} \quad (\because t = 1\text{S})$$

$$\Rightarrow n = 5 \times 10^{15}$$

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Q13

$$(3) \hat{S} = \frac{6\hat{j} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{-3\hat{j} + 4\hat{k}}{5}$$

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Q14

$$(2) B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8} \\ = 20 \times 10^{-8} \text{ T} = 2 \times 10^{-7} \text{ T} \\ K = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi \times 23.9 \times 10^9}{3 \times 10^8} = 500$$

$$\text{Therefore, } \vec{B} = B_0 \sin(kz - \omega t)$$

$$= 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$$

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